



CPU EDUCATION

COMP90051

平时班

□ 4 讲解

TUTOR: JJ 学长

时长: 2 小时

本周内容预览:

1. Model Evaluation (小)
2. 多专家学习+Multiarmbit
3. Bayesian 回归+Bayesian 分类

① NAB UCB 计算 ✓
② NAB 建模

(大) ✓ (小)

(大) (小)



CPU EDUCATION



关于CPU EDUCATION

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课后，如果您有任何建议和意见，我们都非常欢迎您联系小助手分享您的想法，给予我们改进和提高的机会！感谢您参与CPU Education的辅导课程！



课程反馈



PART 1 Model Evaluation

1 Performance Metrics (Pick to fit the task)

Regression. For dataset $D = \{(x_i, y_i)\}_{i=1}^n$ and predictor $g(\cdot)$:

$$\text{MSE}(g) = \frac{1}{n} \sum_{i=1}^n (g(x_i) - y_i)^2, \quad \text{RMSE} = \sqrt{\text{MSE}}, \quad \text{MAE} = \frac{1}{n} \sum_{i=1}^n |g(x_i) - y_i|.$$

Use MAE when outliers are a concern; RMSE penalises large errors more.

Classification. Confusion matrix terms: TP, TN, FP, FN.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}}, \quad \text{Error} = \frac{\text{FP} + \text{FN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}},$$

$$\text{Recall/Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}.$$

Precision and Recall, or Sensitivity and Specificity, often trade off; report them jointly.

ROC & AUC. Sweeping a threshold (or a hyperparameter) traces the ROC curve; AUC summarizes performance (random ≈ 0.5 , perfect = 1).

Asymmetric costs. In health settings, FN vs. FP may differ greatly; consider cost-sensitive loss rather than 0/1 loss.

2 Train/Validation/Test and Cross-Validation

Holdout with validation. Split into train/validation/test. Tune hyperparameters on validation, report final metric on the untouched test set. Repeat/average to reduce variance.

K-fold cross-validation. Partition D into k folds $S_1, \dots, S_k$. For each $i = 1, \dots, k$: train on $D \setminus S_i$, evaluate on S_i , record metric M_i . Report

$$\bar{M} = \frac{1}{k} \sum_{i=1}^k M_i, \quad \widehat{\text{Var}}(M) = \frac{1}{k} \sum_{i=1}^k (M_i - \bar{M})^2.$$

Popular k : 5, 10, or n (LOOCV).

Recipe: k -Fold CV (classifier/regressor)

1. Split indices into k disjoint folds (stratify for class imbalance).
2. For each fold i : fit model on remaining folds, score on fold i .
3. Aggregate mean $\pm$ variability; pick model/hyperparameters accordingly.





3 Avoiding Leakage: Feature Selection & Dimensionality Reduction

Golden rule. Any data-dependent step (scaling, PCA, feature selection) must be fit *only on the training portion* inside each CV split, then applied to that split's validation/test. Never fit on the full dataset before CV.

Data leakage

4 Variability, Error Bars & Hypothesis Testing

Report uncertainty. Do not report only the mean; include variance or standard error (e.g., error bars). Overlapping bars suggest no clear difference.

Two-model comparison via t -test (paired across repeats). Suppose each method is evaluated m times (e.g., k folds or B bootstraps). Let $\hat{\mu}_1, \hat{\mu}_2$ and sample variances $\hat{\sigma}_1^2, \hat{\sigma}_2^2$. Test $H_0 : \mu_1 = \mu_2$ vs. $H_1 : \mu_1 > \mu_2$ with

$$t = \frac{(\hat{\mu}_1 - \hat{\mu}_2)\sqrt{m}}{\sqrt{\hat{\sigma}_1^2 + \hat{\sigma}_2^2}}, \quad \nu = \frac{(\hat{\sigma}_1^2 + \hat{\sigma}_2^2)^2}{\frac{(\hat{\sigma}_1^2)^2}{m-1} + \frac{(\hat{\sigma}_2^2)^2}{m-1}}.$$

Reject H_0 at level α if $t > t_{1-\alpha, \nu}$.

Error bars (significance level α). A common display is $\bar{M} \pm t_{1-\alpha/2, \nu} \cdot \text{SE}$, where SE is the standard error across repeats. Bars that strongly overlap typically indicate no significant difference.

当误差范围重叠较多, 则说明差异不明显

2. (Confusion Matrix) Give formulas for Precision, Recall (Sensitivity), and Specificity.

3. (ROC/AUC) What does an AUC of 0.5 vs. 1.0 indicate? Why can two models with the same accuracy have different AUC?

AUC是通过不同 threshold 下画点连起来

10. (Leakage) Give two common examples of data leakage in CV pipelines and how to fix them.

① PCA in whole data

② scaling in whole data





PART 2 多专家学习+MultiArmbandit

1 Infallible Expert: Majority Vote Algorithm

Assumption: at least one expert is always correct.

Algorithm:

1. Maintain set E of experts not yet wrong.
2. Predict by simple majority of E .
3. Remove experts who err.

Mistake bound:

$$M \leq \log_2 n$$

Each mistake eliminates at least half the experts.

2 Imperfect Experts: Halving Algorithm

Idea: no perfect expert $\Rightarrow$ downweight instead of remove.

1. Initialize weights $w_i = 1$.
2. Predict by weighted majority.
3. If E_i is wrong, update $w_i \leftarrow \frac{1}{2}w_i$.

Bound:

$$M \leq 2.4m + \log_2 n$$

where m is the number of mistakes made by the best expert. This compares learner to the best expert (regret bound form).

3 General Weighted Majority (Multiplicative Weights)

General update:

$$w_i \leftarrow (1 - \varepsilon)^{\ell_i(t)} w_i$$

where $\ell_i(t) \in \{0, 1\}$ indicates expert i 's loss at round t .

Mistake bound:

$$M \leq \boxed{2(1 + \varepsilon)m} + \frac{2 \log_e n}{\varepsilon}$$

Dependence on m improves over Halving algorithm. Constant factor 2 is provably optimal for deterministic methods.





4 Probabilistic Experts Algorithm

Randomized approach:

1. Predict using expert i with probability $p_i = \frac{w_i}{W}$, where $W = \sum_i w_i$.
2. Observe loss $\ell_i(t) \in [0, 1]$.
3. Update: $w_i \leftarrow (1 - \varepsilon)^{\ell_i(t)} w_i$.

Expected loss bound:

$$L \leq \frac{\log_e n}{\varepsilon} + (1 + \varepsilon)L^*$$

where L^* is the loss of the best expert. This removes the deterministic factor of 2.

1 MAB Setting

- Multiple arms with **unknown** reward distributions.
- In each round $t = 1, \dots, T$:

- Play action $i_t \in \{1, \dots, k\}$ (possibly randomly).
- Receive reward $R_{i_t}(t) \sim P_{i_t}$.

- Objective: maximize total expected reward in T rounds (or equivalently, minimize cumulative regret):

$$\text{Regret}(T) = \mu^* T - \sum_{t=1}^T \mathbb{E}[R_{i_t}(t)], \quad \mu^* = \max_i \mu_i.$$

$\Rightarrow \mu^* T$: total reward of the best arm (with hindsight)

$\Rightarrow \sum_t \mathbb{E}[R_{i_t}(t)]$: total actual expected reward.

Explore / Exploit

2 Greedy Algorithm (Pure Exploitation)

- Choose the arm with the largest observed average reward up to round t .

$$Q_{t-1}(i) = \begin{cases} \frac{\sum_{s=1}^{t-1} R_i(s) \mathbb{1}[i_s = i]}{\sum_{s=1}^{t-1} \mathbb{1}[i_s = i]}, & \text{if arm } i \text{ has been played before,} \\ Q_0, & \text{otherwise.} \end{cases}$$

$$i_t \in \arg \max_{1 \leq i \leq k} Q_{t-1}(i)$$

Characteristics:

- Exploits the best arm so far (*no exploration*).
- Highly sensitive to initialization Q_0 .
- May permanently ignore potentially good arms.





3 ϵ -Greedy Algorithm

- Adds exploration to avoid getting stuck in suboptimal arms.

$$Q_{t-1}(i) = \begin{cases} \frac{\sum_{s=1}^{t-1} R_i(s) \mathbb{1}[i_s = i]}{\sum_{s=1}^{t-1} \mathbb{1}[i_s = i]}, & \text{if arm } i \text{ has been played before,} \\ Q_0, & \text{otherwise.} \end{cases}$$

$$i_t \sim \begin{cases} \arg \max_{1 \leq i \leq k} Q_{t-1}(i), & \text{with prob. } 1 - \epsilon; \\ \text{Uniform}\{1, \dots, k\}, & \text{with prob. } \epsilon. \end{cases}$$

我们 Greedy 策略

Interpretation:

- Exploitation with probability $1 - \epsilon$.
- Exploration (random arm) with probability ϵ .
- ϵ can be fixed (e.g. 0.1) or decay over time (e.g. $\epsilon_t = 1/t$).

前面 round 越大
后面越慢

- When exploring, choice is blind—it does not consider confidence or potential.

4 Initialization Strategy (Cold Start Handling)

Goal: adjust Q_0 to manage the exploration-exploitation trade-off.

- Pessimistic Initialization:** small Q_0 (e.g. $Q_0 = 0$) $\Rightarrow$ encourages exploitation.
- Optimistic Initialization:** large Q_0 (e.g. $Q_0 = 1000$) $\Rightarrow$ encourages exploration.
- Middle-ground:** Q_0 moderate $\Rightarrow$ each arm explored about once.

Key difference:

- Works only in the initial stage (cold start).
- Does not adapt dynamically over time.

Q_0 fixed setting

$$Q_0 = 0$$

1	2	3
0	0	10

$$\frac{10}{2} = 5$$

$$Q_0 = 1000$$

1	2	3
1000	1000	1000

$$\frac{1000+1000}{2} = 1005$$





5 Upper Confidence Bound (UCB)

大起快并

Idea: "Optimism in the face of uncertainty" — dynamically balance exploration and exploitation.

多 explore 那一个

代表着对 uncertainty 的奖励

$$Q_{t-1}(i) = \begin{cases} \hat{\mu}_{t-1}(i) + \sqrt{\frac{\rho \log t}{N_{t-1}(i)}}, & \text{if } N_{t-1}(i) > 0, \\ Q_0, & \text{otherwise.} \end{cases}$$

$$i_t \in \arg \max_{1 \leq i \leq k} Q_{t-1}(i)$$

Where:

高起的 initialization

- $\hat{\mu}_{t-1}(i)$ = average reward of arm i up to round $t-1$.
- $N_{t-1}(i)$ = number of times arm i has been played so far.

- ρ = exploration coefficient (controls the exploration strength).

越大越鼓励 explore

Mechanism:

- Larger $\log t$ term $\uparrow$ $\Rightarrow$ gradually decreases exploration as t increases.
- Rarely played arms ($N_{t-1}(i)$ small) receive a larger bonus.
- Eventually converges to optimal arm as uncertainty decreases.

log T 增长不如 $N_t(i)$ 所以奖励项随着轮次
的作用越来越小

Regret:

$$\mathbb{E}[\text{Regret}(T)] = O(\log T)$$

— asymptotically optimal up to a constant factor.

6 Comparison: Initialization vs. UCB

Method	When It Works	Limitation
Reward Initialization	Only at start (cold start); depends on Q_0	Stops exploring once arms have been sampled
UCB	Continuous adaptation for all t	Needs ρ tuning; may over-explore early

Greedy, E-Greedy
Explore

Explore

一般:
 $\rho=2$ 

**Question 17: Multi-Armed Bandits [8 marks]**

Describe two mechanisms used by the Upper Confidence Bound (UCB) multi-armed bandit (MAB) to explore arms that so far have no demonstrated high rewards.

The following formulae may be of use:

UCB arm i value estimate after observing rewards: $Q_{t-1}(i) = \hat{\mu}_{t-1}(i) + \sqrt{\frac{\rho \log(t)}{N_{t-1}(i)}}$ where $N_{t-1}(i)$ is the number of observations of arm i , $\hat{\mu}_{t-1}(i)$ is the average of observed rewards on arm i so far; $\rho > 0$ a hyperparameter. Unobserved arms are initialised with value estimate Q_0 a hyperparameter.

① Exploration Bonus for arms with uncertainty

the $\sqrt{\frac{\rho \log(t)}{N_{t-1}(i)}}$ is big if $N_{t-1}(i)$ is small, means it hasn't been tried a lot, so the algorithm will give a large bonus to it even if there was no demonstrated high reward

② Optimistic Initialization, with a higher Q_0 , it will encourage arms to be explored more

- (a) Describe where within multi-armed bandit algorithms like ϵ -greedy, statistical estimation is performed as a subroutine. Describe how these estimates are used. [2 marks]

Empirical mean reward for each arm

Estimates used to guide arm selection

we choose the arm with highest mean reward

- (c) Both the *learning with experts* and multi-armed bandit settings involve learning to optimise cumulative rewards (or equivalently losses). What is a key difference between the settings? [5 marks]

Experts: learners receive all feedback after each round

MAB: only receive partial feedback
(reward of the selected arm)

Question 4: Multi-Armed Bandits [2 marks]

Describe two mechanisms used by the Upper Confidence Bound (UCB) multi-armed bandit (MAB) to explore arms that so far have no demonstrated high rewards. You may refer to the formula sheet for specifics of UCB.





Question 2: [6 marks]

✓✓✓ 去年s2考试

You are running a multi-armed bandit problem with 3 arms. The rewards from each arm are binary (0 or 1), and so far, you have pulled the arms with the following results:

- Arm 1: Pulled 64 times, with 40 rewards of 1.
- Arm 2: Pulled 16 times, with 10 rewards of 1.
- Arm 3: Pulled 4 times, with 2 reward of 1.

Using the UCB algorithm, calculate the value of each arm based on the given results. You can provide your answer in fractions or round your answers to 3 decimal places. Based on your calculations, which arm should you pull next, and why?

Some possibly useful values: $\ln(4) \approx 1.4, \ln(5) \approx 1.6, \ln(16) \approx 2.8, \ln(17) \approx 2.8, \ln(64) \approx 4.2, \ln(65) \approx 4.2, \ln(84) \approx 4.5, \ln(85) \approx 4.5$

Some possibly useful formulas:

t: 当前轮次

$$Q_{t-1}(i) = \hat{\mu}_{t-1}(i) + \sqrt{\frac{2 \ln t}{N_{t-1}(i)}} \quad \text{where } N_{t-1}(i) = \sum_{s=1}^{t-1} 1[i_s = i], \quad \hat{\mu}_{t-1}(i) = \frac{\sum_{s=1}^{t-1} R_i(s) 1[i_s = i]}{N_{t-1}(i)}$$

Total pulls: $64 + 16 + 4 = 84$
 当前 round $84 + 1 = 85$ $\ln 85 \approx 4.5$

$$Q_{t-1}(i) = \hat{\mu}_{t-1}(i) + \sqrt{\frac{2 \ln t}{N_{t-1}(i)}}$$

first term =

Arm 1: $\hat{\mu}_{t-1}(1) = \frac{40}{64} = 0.625$ $N = 64$

Arm 2: $\hat{\mu}_{t-1}(2) = \frac{10}{16} = 0.625$ $N = 16$

Arm 3: $\hat{\mu}_{t-1}(3) = \frac{2}{4} = 0.5$ $N = 4$

Second term =

Arm 1: $\sqrt{\frac{9}{64}} = 0.375$

Arm 2: $\sqrt{\frac{9}{16}} = 0.75$

Arm 3: $\sqrt{\frac{9}{4}} = 1.5$

Arm 1: $0.625 + 0.375 = 1$

Arm 2: $0.625 + 0.75 = 1.375$

Arm 3: $0.5 + 1.5 = 2$

choose arm 3

$Q_t(3)$ 是最大的





You are testing a multi-armed bandit algorithm with 3 arms. The rewards from each arm are binary (0 or 1). So far, you have pulled the arms as follows:

- Arm 1: Pulled 25 times, with 15 rewards of 1.
- Arm 2: Pulled 10 times, with 6 rewards of 1.
- Arm 3: Pulled 5 times, with 3 rewards of 1.

$$\hat{\mu}_1 =$$

Using the **Upper Confidence Bound (UCB)** algorithm, calculate the value of each arm based on the given results. You may provide your answer in fractions or round to **3 decimal places**. Based on your calculations, which arm should be pulled next, and why?

Some useful values: $\ln(5) \approx 1.6$, $\ln(10) \approx 2.3$, $\ln(25) \approx 3.2$, $\ln(40) \approx 3.7$, $\ln(41) \approx 3.7$.

UCB formula:

$$Q_{t-1}(i) = \hat{\mu}_{t-1}(i) + \sqrt{\frac{2 \ln t}{N_{t-1}(i)}}$$

where $\hat{\mu}_{t-1}(i)$ is the average reward for arm i , and $N_{t-1}(i)$ is the number of times arm i has been pulled.

未#, Answer:

$$\text{Total pulls} = 25 + 10 + 5 = 40$$

$$T = 41 \quad \ln 41 \approx 3.7$$

second term

$$\hat{\mu}_1 = \frac{15}{25} = 0.6$$

$$\hat{\mu}_2 = \frac{6}{10} = 0.6$$

$$\hat{\mu}_3 = 0.6$$

$$\sqrt{\frac{2 \cdot 3.7}{25}}$$

$$\sqrt{\frac{2 \cdot 3.7}{10}}$$

$$\sqrt{\frac{2 \cdot 3.7}{5}}$$

由于 $\hat{\mu}_i$ 一样, Arm 3 拉最少, second term 最大
所以选择 Arm 3 Next



**Question 6: Online Content Testing [20 marks]**

Your task is to design an automatic *content recommendation* system for a major newspaper "TechNews" to use to select which headlines to use for articles on its online front page, and which pictures to display for these front page article previews. TechNews has very many *users*, and serves very many different *articles* across different categories (like local news, international news, opinion pieces, investment, etc.). TechNews wants to encourage user engagement with their platform, so that users keep returning to the news service (driving more subscription and ad revenue). TechNews management therefore will judge the performance of its content recommendation system by cumulative clicks. Complicating matters, different kinds of users typically click on different kinds of articles according to their tastes in article categories, types of headlines, and picture previews, etc. Your goal is to recommend headlines and pictures to display to *maximise* the total number of clicks, where each article comes with a number of different available headlines and picture previews to choose from.

加入了环境

- Formulate the *content recommendation* task as a contextual multi-armed bandit problem. Explain, what your *arms* are, what *rounds* correspond to, and what are your *rewards*. [6 marks]
- What context vectors will you use based on what data TechNews might possess? [6 marks]
- Comment on whether the typical bandit assumption of independent and identically distributed rewards per arm is appropriate. Does context make this assumption more or less realistic? [4 marks]
- Describe what data TechNews should *collect*, so that you could *efficiently evaluate* the effectiveness of a huge number of bandit learners. [4 marks]

a). Round: Each time user load the Technews frontpage

Arms: Candidate headlines and pictures

Reward: click or not

b). context: users feature: location, device, age
article feature: category

c). 现实中 reward 往往不是 iid, 比如用户过去该发它什么类题材
作为给定 context, reward 更接近 iid

d). round id
Each chosen article
outcome
context feature



**Question 6: Surveillance in a Retail Store [20 marks]**

Your task is to design a system to optimise the *placement of a fake security camera* in a small retail store. The fake camera is large and easily seen by nearby customers. Depending on where the fake camera is placed, shoppers will be discouraged or deterred from stealing different items from the store. There are 10 locations the fake camera may be placed, and the store doesn't know which is best. The store would like you to design a system that decides which of the 10 locations to move the fake camera every day, to minimise total cumulative thefts in the store. You can assume the store performs a full inventory daily to accurately measure total theft.

- Considering the 10 locations as 10 possible actions per day, discuss what you could use as *rewards* or *losses* per day, that could be suitable either for a bandit or learning with experts solution. [4 marks]
- Suppose the fake camera is left in the same position over multiple days. To what extent are the daily rewards *independent*? [4 marks]
- Assuming independent rewards, Which setting is more appropriate for this problem: *multi-armed bandits* or *learning with experts*? (Where locations are either arms/actions or experts.) In your answer, make reference to whether in one day, *rewards are observed* for only the used fake camera location, or all locations. [6 marks]
- The store now wants to place a *real camera* in one of 10 locations. You may decide on where to place the camera, each day. Assume that the *real camera* is very small and invisible to customers. Assume the real camera can be used to accurately *identify thieves* nearby to the chosen location. To maximise the number of cumulative thieves identified, should you use *bandit learning* or *learning with experts*? Why. What rewards/losses would you use? [6 marks]

a). Round: Each day
losses: total theft of that day
Arms: choose one of 10 location

b). No, 小偷可能会先学习 camera 位置, 这样后面的 theft 和前面形成了联系

c). Partial feedback, Bandit setting

d).





X Question 12: Online Media Streaming [5 marks]

Your task is to design an automatic *thumbnail recommendation* system for Netflix to use to select which small preview icons to display for each movie and TV show it lists in user homepages. Netflix has very many *users*, and serves very many different *movies and shows* across different genres (like science fiction, comedy, romance, drama, etc.) Because Netflix wants to encourage user engagement with their platform, so that users keep subscribing to the for-fee streaming service, Netflix management will judge the performance of its thumbnail recommendation system by *cumulative clicks*. Complicating matters, different kinds of users typically click on different kinds of shows according to their tastes in genres, favourite actors, etc. Your goal is to *recommend* thumbnails to display to *maximise* the total number of clicks, where each show comes with a number of different available thumbnail graphics to choose from.

- (a) Formulate the *thumbnail recommendation task* as a *contextual multi-armed bandit problem*. Explain, what your arms are, what rounds correspond to, and what are your rewards. [1 mark]
- (b) What *context* vectors will you use based on what data Netflix might possess? [2 marks]
- (c) Comment on whether the typical bandit assumption of *independent and identically distributed* rewards per arm is appropriate. Does context make this assumption more or less realistic? [1 mark]
- (d) Describe what data Netflix should *collect*, so that you could *evaluate* the effectiveness of multiple bandit learners you might try. [1 mark]

a). Round: 每次用户打开 home page

Arms: the alternative thumbnails

reward: user click

b).

user: age, 地区, 时间

内容: 内容类别, 导演

c). 同一个用户喜好可能与前几天的推荐也有关, 所以不能

仅引入 3 context, 分布更稳定, iid 更合理

d).

userId

选中哪条 arm

context
outcome (reward)





Question 8: COVID-19 Vaccine Selection [18 marks]

Many months have passed since the onset of COVID-19, and 5 medical laboratories have created and trialed 5 vaccines with no significant adverse side effects after careful clinical trials. From the trials it is believed that some vaccines are more effective at preventing contraction of COVID-19 for some patients, but no one knows which. Doctors across the world must now determine *how to select the right vaccine for any given patient*, as patients stream into general practice clinics. How might you as Chief Health Officer use machine learning to help fight COVID-19 by *recommending appropriate vaccines*?

- Formulate the *vaccine selection task* as a non-contextual *multi-armed bandit problem*. Explain what your arms are, what rounds correspond to, and what are your rewards. [6 marks]
- Suppose in earlier clinical trials various measurements were taken of patients prior to test vaccination. Given access to these measurements, and effectiveness of the vaccinations in trials, how might you decide whether *contextual multi-armed bandits* might offer an improvement over your non-contextual solution? [4 marks]
- Suppose 100 machine learning companies approach you with their MAB solutions, that have never been previously tested. What drawback is there to trying all of the 100 approaches out on real patients in the wild? [4 marks]
- Now suppose the prior clinical trials were completely randomised, such that for a patient that arrived, a *uniformly random* vaccine was given to them, and the result recorded. How could you use this data to *evaluate all 100 MAB learners*? [4 marks]

a). Arm: 每种 vaccine
round: 自来一病人, 选择一疫苗
Reward: 该病人是否能避免被感染

b). 已经有病人数据, contextual 更好

c). 测试完全都需要很多 trials, 对病人可能有副作用

d). Assume Data format

	patient	vaccine	reward
✗ 不是	1	2	0
	2	4	0
	1	4	1

假设非 MAB

推荐 Patient 1 vaccine 4

因为 MAB 推荐的是 vaccine 4

只有这条 record 加入





Question 6: Dynamic Product Pricing [20 marks]

Your task is to design an automatic *price recommendation* system for a major snack food maker “Acme” (a fictitious company) to use to select which prices to sell its products for. Acme make a range of low-to high-end snacks, including potato chips/crisps, chocolates, boiled lollies, cookies, and more. Acme sells its many *products* in supermarkets across many *countries*. Naturally Acme want to maximise their *profit*, by carefully setting their product *pricing* — the amount it will sell its products for (assume the same ‘Acme dollars’ currency everywhere). Acme can set a price per product per country, and vary this choice over time. Priced too low, Acme may lose profit per item or even make a loss on a product; priced too high and Acme may sell too few items, or even none at all. Acme management will judge the performance of its price recommendation system by *cumulative profit*. Complicating matters, people in different countries might have different tastes for Acme products, and different amounts they’re willing to pay. Your goal is to *recommend* prices to set to *maximise* the total profit of Acme.

- Formulate the *price recommendation task* as a *contextual multi-armed bandit problem*. Explain, what your *arms* are, what *rounds* correspond to, and what are your *rewards*. [6 marks]
- What *context* vectors could you use based on what data Acme might possess? [4 marks]
- Comment on whether the typical bandit assumption of *independent and identically distributed* rewards per arm is appropriate. [4 marks]
- Consider now using a *non-contextual bandit*. How would you change your design (if at all)? And how might the use of non-contextual bandits affect the amount of *exploration* needed, and speed of *convergence* to finding good prices? [6 marks]

a). arm: 一系列价格
round: 每次定价决策
rewards: Profit

b). 产品分类: 国家, 季节

c). 现实中, 很多 context 都影响 Profit
比如 消费行为, 前一天买了 chocolate
第二天没吃完不买

d). Non-context (无国家之分)

只能全局运用 Bandit

因为全局使用 arms 更多, 更多 exploration
收敛速度更慢





PART 3 贝叶斯回归和贝叶斯分类

The Bayesian Learning Process

- Start with a prior belief about parameters θ : $p(\theta)$.
- Collect data $\mathcal{D} = \{x_i, y_i\}$ and write the likelihood: $p(\mathcal{D}|\theta)$.
- Apply Bayes' rule:

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})} \propto p(\mathcal{D}|\theta)p(\theta)$$

- The posterior combines prior knowledge and evidence.
- Predict for new input x^* by marginalising over uncertainty:

$$p(y^*|x^*, \mathcal{D}) = \int p(y^*|x^*, \theta) p(\theta|\mathcal{D}) d\theta$$

① 找 Prior

② 找 likelihood

③ 用 Bayesian EL

④ 求后验

$p(\theta|\mathcal{D})$ — MAP 估计
先验期望
抽样

Understanding Conjugacy

- A conjugate prior means the posterior is in the same family as the prior.

$$p(\theta|\mathcal{D}) \propto p(\mathcal{D}|\theta)p(\theta) \Rightarrow p(\theta|\mathcal{D}) \text{ has same form as } p(\theta)$$

- Advantage: easy analytical updates; just update hyperparameters.
- Rule of thumb: drop constants not involving θ and match the functional form.

Beta(α, β)

conjugate 可以不计算 evidence
来求 posterior

Example: Beta-Binomial (Coin Toss)

$$\begin{aligned} p(q|\text{data}) &\propto q^k(1-q)^{n-k} \times q^{\alpha-1}(1-q)^{\beta-1} \\ &= q^{\alpha+k-1}(1-q)^{\beta+n-k-1} \\ &\Rightarrow q|\text{data} \sim \text{Beta}(\alpha+k, \beta+n-k) \end{aligned}$$

Posterior mean:

$$\mathbb{E}[q|\text{data}] = \frac{\alpha + k}{\alpha + \beta + n}$$





Bayesian Linear Regression (Normal-Normal)

$$y = \mathbf{x}^\top \mathbf{w} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \alpha^{-1} \mathbf{I})$$

Posterior:

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \mathcal{N}(\mathbf{w}_N, \mathbf{V}_N)$$

where

$$\mathbf{V}_N = (\alpha \mathbf{I} + \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X})^{-1}, \quad \mathbf{w}_N = \frac{1}{\sigma^2} \mathbf{V}_N \mathbf{X}^\top \mathbf{y}.$$

Predictive distribution:

$$p(y^*|\mathbf{x}^*, \mathcal{D}) = \mathcal{N}(\mathbf{x}^{*\top} \mathbf{w}_N, \sigma^2 + \mathbf{x}^{*\top} \mathbf{V}_N \mathbf{x}^*).$$

Bayesian Classification 两种视角

Generative View (Naïve Bayes-style)

$$p(y|\mathbf{x}) \propto p(\mathbf{x}|y)p(y)$$

Each class y has its own likelihood $p(\mathbf{x}|y)$, and prediction picks $\arg \max_y p(y|\mathbf{x})$.

Discriminative View (Bayesian Logistic Regression)

$$p(y=1|\mathbf{x}, \mathbf{w}) = \sigma(\mathbf{w}^\top \mathbf{x}), \quad p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \alpha^{-1} \mathbf{I})$$

No closed-form posterior: $\Rightarrow$ non-conjugate.

Approximation methods:

- Laplace: Normal approximation around MAP.
- Variational: optimise a tractable $q(\mathbf{w})$ to match posterior.
- Monte Carlo: draw samples and estimate expectations.



**Question 9: Bayesian Conjugacy [4 marks]**

This question explores *Bayesian conjugacy*, relating to the following distribution:

Inverse Gamma distribution. $y \sim \text{IGamma}(\alpha, \beta)$ has probability density $p(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{-\alpha-1} \exp\left(-\frac{\beta}{y}\right)$ for variable y taking any positive real value and parameters shape α and scale β both taking any positive real values. ($\Gamma(\alpha)$ is the Gamma special function that comes from normalisation and is just a constant.)

- (a) Prove that an Inverse Gamma on σ^2 is a conjugate prior for a Gaussian likelihood on x with known μ and unknown variance parameter σ^2 . (Hint: the density for prior on σ^2 is found by setting $y = \sigma^2$ above.) [3 marks]
- (b) You observe independent $X_1, \dots, X_n$ drawn from Gaussian of known μ with prior $\text{IGamma}(\alpha, \beta)$ on unknown σ^2 . What are the posterior updated Inverse Gamma hyperparameters depending on $\alpha, \beta, X_1, \dots, X_n$? If unable to answer the previous part, you may assume a function for the posterior update first. [1 mark]

$$a). \text{ prior: } p(\sigma^2) = \frac{\beta^\alpha}{\Gamma(\alpha)} (\sigma^2)^{-\alpha-1} \exp\left(-\frac{\beta}{\sigma^2}\right) \checkmark$$

$$\text{likelihood: } \prod_{i=1}^n p(x_i | \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

posterior $\propto$ likelihood $\times$ Prior

$$\propto (\sigma^2)^{-\alpha-1} \cdot (\sigma^2)^{-\frac{n}{2}} \cdot \exp\left(-\frac{\beta}{\sigma^2} - \frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$= (\sigma^2)^{-(\alpha+0.5)} \cdot \exp\left(-\frac{1}{\sigma^2} \left(\beta + \frac{(x-\mu)^2}{2}\right)\right)$$

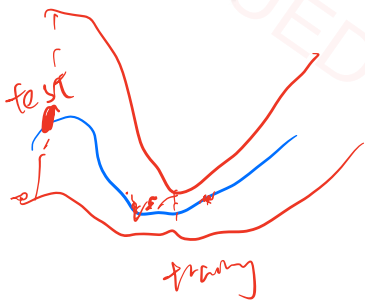
$$\text{posterior} \sim \text{IGamma}(\alpha+0.5, \beta + \frac{(x-\mu)^2}{2})$$

11. For *Bayesian linear regression* (as presented in class) explain how the posterior variance can vary for different test points. [4 marks]

The following formulae may be of use:

Bayesian regression with prior $\mathbf{w} \sim \text{Normal}(0, \gamma^2 \mathbf{I}_D)$ and likelihood $y \sim \text{Normal}(\mathbf{x}^\top \mathbf{w}, \sigma^2)$ leads to a posterior of the form $\mathbf{w} \sim \text{Normal}(\mathbf{w}_N, \mathbf{V}_N)$ and Gaussian predictive distribution

$$y_* \sim \text{Normal}(\mathbf{x}_*^\top \mathbf{w}_N, \sigma^2 + \mathbf{x}_*^\top \mathbf{V}_N \mathbf{x}_*), \text{ where } \mathbf{w}_N = \frac{1}{\sigma^2} \mathbf{V}_N \mathbf{X}^\top \mathbf{y} \text{ and } \mathbf{V}_N = \sigma^2 \left(\mathbf{X}^\top \mathbf{X} + \frac{\sigma^2}{\gamma^2} \mathbf{I}_D \right)^{-1}.$$



$$\text{Var}(y^*) = \sigma^2 + \mathbf{x}_*^\top \mathbf{V}_N \mathbf{x}_*$$

方差是依赖于 Data points, 就是对 $\mathbf{w}$ 的不确定性

当 test point 靠近训练点, variance 更小

当 test point 远离训练点, variance 更大



Q9. b.

$X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$

$$p(\sigma^2 | x) \propto (\sigma^2)^{-\alpha+1} (\sigma^2)^{-\frac{m}{2}} \exp\left(-\frac{\beta}{\sigma^2} - \frac{\sum_{i=1}^m (x_i - \mu)^2}{2\sigma^2}\right)$$
$$= (\sigma^2)^{-(\alpha + \frac{m}{2}) - 1} \exp\left(-\frac{1}{\sigma^2} \left(\beta + \frac{\sum_{i=1}^m (x_i - \mu)^2}{2}\right)\right)$$

$$\alpha_{\text{new}} = \alpha + \frac{m}{2}$$

$$\beta_{\text{new}} = \beta + \frac{\sum_{i=1}^m (x_i - \mu)^2}{2}$$



Show that Inverse Gamma is the conjugate prior for the variance in Normal distribution. The probability density function for random variable y of the inverse gamma distribution is:

$$g(y; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{-(\alpha+1)} e^{-\frac{\beta}{y}} \quad \text{Prior}$$

where α and β are parameters, $\Gamma(\alpha)$ is a function about α .

The probability density function for random variable x of the normal distribution is:

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{Likelihood}$$

where μ and σ are parameters.

$$\text{Prior: } g(b^2; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot (b^2)^{-(\alpha+1)} \cdot e^{-\frac{\beta}{b^2}}$$

$$\begin{aligned} \text{Likelihood: Assume } x_1, \dots, x_n \\ \prod_{i=1}^n f(x_i | \mu, b^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}b} \cdot e^{-\frac{(x_i - \mu)^2}{2b^2}} \\ &\propto (b^2)^{-\frac{n}{2}} \cdot e^{-\frac{1}{2b^2} \sum_{i=1}^n (x_i - \mu)^2} \\ &= (b^2)^{-\frac{n}{2}} \cdot e^{-\frac{S}{2b^2}} \\ S &= \sum_{i=1}^n (x_i - \mu)^2 \end{aligned}$$

$$\begin{aligned} \text{Posterior} &\propto \text{Likelihood} \times \text{Prior} \\ p(b^2 | x) &\propto (b^2)^{-\frac{n}{2}} \cdot e^{-\frac{S}{2b^2}} \cdot (b^2)^{-(\alpha+1)} \cdot e^{-\frac{\beta}{b^2}} \\ &= (b^2)^{-(\alpha + \frac{n}{2} + 1)} \cdot e^{-\frac{1}{b^2}(\beta + \frac{S}{2})} \end{aligned}$$

Inverse Gamma

$$\left(\alpha + \frac{n}{2}, \beta + \frac{S}{2} \right)$$













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